

Education

- 2025–Present Montana State University—M.S. Physics, expected 2027
Graduate Research Assistant, Jaffe Lab
- 2021–2025 University of Idaho—B.S. Physics

Research

Graduate Research Assistant — Jaffe Lab, Montana State University

2025–Present

- Design, construct, align, and characterize free-space optical systems for research in programmable photonics and quantum networking.
- Develop experimental testbeds incorporating polarization optics, electro-optic modulation, interferometry, photodetection, and scientific imaging.
- Perform optical characterization using power, beam-profile, polarization, and camera-based measurements. Develop Python tools to analyze measurements and compare experimental behavior with physical models.
- Contribute to programmable photonics research with applications in optical communication and quantum networking.

Undergraduate Research Assistant — Vasdekis Lab, University of Idaho

2024–2025

- Investigated applications of optical thermodynamics to imaging and non-Gaussian beam propagation in TIRF microscopy.
- Constructed and aligned free-space optical experiments and performed beam-profile and optical-power measurements.
- Assisted with experimental characterization and analysis of structured and non-Gaussian optical fields.

Research Assistant — Etienne Numerical Relativity Group, University of Idaho

2021–2025

- Developed Python and Jupyter workflows for processing and visualizing numerical-relativity and GRMHD simulation data.
- Extended visualization tools for binary black-hole simulations and collaborated on improvements to code performance, maintainability, and reproducibility.
- Developed and documented a reusable framework for analysis and visualization of large scientific datasets.

Technical Skills

Experimental Optics:

Free-space optical alignment, polarization optics, interferometry, optical power measurement, beam profiling, and scientific camera imaging

Optical Instrumentation:

Electro-optic modulators, liquid-crystal variable retarders, photodiodes, optical power meters, beam profilers, and piezoelectric actuators

Experimental Computing:

Python-based data analysis and visualization, camera acquisition, and basic instrument control

Software:

Python, NumPy, SciPy, Matplotlib, Jupyter, Git, and Linux/Unix

Prototyping:

3D printing, custom optomechanical components, and optical breadboard assembly

Affiliations

2024–Present	Student Member , Institute of Electrical and Electronics Engineers (IEEE)
2024–Present	Student Member , Society of Photo-Optical Instrumentation Engineers (SPIE)